AGGREGATE FUNCTIONS

INTRODUCTION

An aggregate function performs a calculation on a set of values of a column, and returns a single value

Built-in aggregate functions

- **`COUNT`**, **`SUM`**, **`MAX`**, **`MIN`**, and **`AVG`**

Functions can be used in the `SELECT` clause or in a `HAVING` clause

EXAMPLE

Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary.

```
SELECT    SUM (Salary), MAX (Salary), MIN (Salary), AVG(Salary)
FROM      EMPLOYEE;
```

Sum (Salary)	Max(Salary)	Min (Salary)	Avg (Salary)
986550	55000	25000	55600

EXAMPLE

Find the total number of employees.

```
SELECT        count(ssn)
FROM     EMPLOYEE;
```

Count(ssn)
8

The above query will ignore nulls but not duplicates.

```
SELECT        count(*)
FROM     EMPLOYEE;
```

Count(*)
8

The above query will not ignore nulls & duplicates. count(*) counts the rows.

HOW TO REMOVE DUPLICATES FROM RESULT OF COUNT?

Count the number of distinct salary values in the database

SELECT **COUNT** (**DISTINCT** Salary)

FROM EMPLOYEE;

Output = 6

GROUP BY

Aggregate functions are often used with the GROUP BY clause

GROUP BY Clause is used to collect data from multiple records and group the result by one or more column

Example: "find the number of employees in each department".

- Select count(*), dno

From employee

Group by dno;

Count(*)	Dno
1	1
3	4
4	5

GROUP BY

Introduction to MySQL GROUP BY clause

The `GROUP BY` clause groups a set of rows into a set of summary rows by values of columns or expressions. The `GROUP BY` clause returns one row for each group. In other words, it reduces the number of rows in the result set.

The `GROUP BY` clause is an optional clause of the `SELECT` statement. The following illustrates the `GROUP BY` clause syntax:

```
SELECT
    c1, c2,..., cn, aggregate_function(ci)
FROM
    table
WHERE
    where_conditions
GROUP BY c1 , c2,...,cn;
```

GROUP BY

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GROUP BY Clause is used to collect data from multiple records and group the result by one or more column

Example: "find the number of employees in each department".

- Select count(*), dno

From employee

Group by dno;

Count(*)	Dno
1	1
3	4
4	5

GROUP BY SUBJECT COUNT ? GROUP BY YEAR COUNT?

SUBJECT	YEAR	NAME
English	1	Harsh
English	1	Pratik
English	1	Ramesh
English	2	Ashish
English	2	Suresh
Mathematics	1	Deepak
Mathematics	1	Sayan

SYNTAX

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

find the number of employees in each department

Select dno, count (*)
From employee
Group by dno ;

Dno	Count(*)
4	3
5	4
1	1

EXAMPLE:

For each department, retrieve the department number, the number of employees in the department, and their average salary.

SELECT Dno, **COUNT** (*), **AVG** (Salary)
FROM EMPLOYEE
GROUP BY Dno;

Dno	Count(*)	Avg(Salary)
4	3	30000
5	4	31000
1	1	55000

PRACTICE:

For each project, retrieve the project number, the project name, and the number of employees who work on that project.

Select pnumber, pname, count(essn)

From workson join project on pno = pnumber

Group by pnumber,pname

GROUP BY EXAMPLE

For each project, retrieve the project number, the project name, and the number of employees from department 5 who work on the project.

Select pnumber, pname, count(essn)

From (project join workson on pno= pnumber) join employee on essn =ssn

Where dno = 5

group by pnumber, pname;

HAVING CLAUSE

Used for condition (just like where clause) on aggregate function's result.

Having clause will not execute if there is no group by.

HAVING CLAUSE

The MySQL HAVING Clause

The HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions

EXAMPLE

Show department names which have more than two locations.

Select dname, count(dlocation) as total_locations

From department join dept_locations using (dnumber)

Group by dname

Having count(dlocation) >2

Order by dname ;

ORDER OF EXECUTION

ORDER	CLAUSE	FUNCTION
1	from	Choose and join tables to get base data.
2	where	Filters the base data.
3	group by	Aggregates the base data.
4	having	Filters the aggregated data.
5	select	Returns the final data.
6	order by	Sorts the final data.
7	limit	Limits the returned data to a row count.

PRACTICE QUERIES (CP)

1. Show average salary of all employees.
2. Show total number of departments.
3. Show sum of hours spent on all projects.
4. Show average salary of each department if average salary is greater than 40,000.
5. Show minimum salary of each department if minimum salary is less than 20,000.
6. Show total number of employees of each department with department name.
7. Show number of dependent of each employee.
8. Show count of projects of all departments in ascending order by department name.
9. Show count of total locations of each department which are located in Lahore.
10. For each project show project name and total number of employees working on it.